A Fast Algorithm for Optimal Power Scheduling of Large-Scale Appliances With Temporally Spatially Coupled Constraints

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***Abstract*—Optimally scheduling power consumption of appliances is the essential feature of smart grid, which enables Demand Response Management (DRM) and helps to shape the power usage profile. This problem is often required to be solved in face of a large number of appliances and many time slots; thus the computational efficiency of solving a large scale optimal power scheduling problem with limited computational resources becomes the major concern of algorithm design. To this end, a novel algorithm is proposed based on Karush-Kuhn-Tucker (KKT) conditions to solve the optimal power scheduling problem with temporally spatially coupled constraints in a distributed manner. The proposed algorithm converts the original problem into equivalently solving an optimal KKT operator in a much lower dimension, thus the computation speed is greatly enhanced. In addition, the proposed method dose not require a step size in the iteration process, thus avoids the oscillation of numerical solution caused by problem parameter changes. Compared with the widely used conventional algorithms, e.g., interior point method and dual decomposition, the higher computational speed and less sensitivity to the problem parameter setting are observed in numerical simulations.**

***Index Terms*—Appliances power consumption scheduling, computational efficiency, KKT conditions, dual decomposition.**

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Nomenclature

Functions

S	Derivative function of W'^{-1}
W	Diviation cost function
W'	Derivative function of W
W'^{-1}	Inverse function of W'.

Numbers and Indices

A	Cardinal number of set $\mathcal{A}$
a	Indices for appliances
H	Cardinal number of set $\mathcal{H}$
h	Indices for time slots.

Parameters

α	Electricity price
$\overline{P}$	Physical power limit of the district transformer or substation transformer
$\overline{p}$	Constant rated power
D	Total energy needed for a task
o	Weighting factor
p^{max}	Maximum power limit
p^{min}	Minimum power limit
T	Duration of a time slot.

Sets and Vectors

$\mathcal{A}'$	Set of appliances whose power are below the maximum
$\mathcal{A}$	Set of shiftable appliances
$\mathcal{H}$	Workable time slots
P	Power vector of the appliance.

Variables

Δp	Deviation of actual power and rated power
p	Energy power.

I. Introduction

By having smart meters installed at users' premises and two-way communications enabled between the power utilities and users, Demand Response Management (DRM) becomes a fundamental feature of smart grid. DRM is the response of end users in power usage to changes in electricity prices over time or to other forms of incentive, aiming at improving energy efficiency and reducing cost [2]–[5]. In practice, DRM is realized by optimally scheduling the power of controllable appliances with the objective to reach a cost minimum. In general, optimal power scheduling problem can be written into a mathematical programming problem subject to kinds of constraints.

In order to efficiently solve the power scheduling problem of appliances, many advanced methods have been designed. Distributed methods are emerging as a primary way to address this problem, which can be classified into three categories. The first category of methods models the problem as a convex optimization problem and propose algorithms to solve the problem in a distributed manner. Optimization techniques, such as, [6] KKT conditions, [7], [8] dual decomposition method and Alternating Direction Method Of Multipliers (ADMM) [9], are widely used in these studies. The second category of papers includes more realistic conditions that treat the problem as non-convex problem, and tailored methods are proposed for distributed solution. In [10], a bidirectional framework is developed in a distributed way and a Newton method is employed to accelerate convergence. In [11], the authors present a fast distributed algorithm, which is applied to the double smoothed dual function of the problem. In [12], a suboptimal distributed algorithm based on an extended Lyapunov optimization technique is used. In [13], an approximate greedy iterative algorithm is used to find suboptimal solution. The third type of problems are using concepts from game theory to solve the problem distributively, which generally converge to Nash or other equilibriums. The scheduling problem is modeled, separately, as a coupled-constraint game in [14], and a non-cooperative game in [15]. Reference [16] proposes a game theoretic approach based on a modified regret matching procedure, and minimizing the energy costs is achieved at the Nash equilibrium in [17].

Often, the power scheduling problem is further complicated by two types of constraints: 1) temporal constraint that enforces an appliance to consume a certain amount of electricity in order to complete the given task within the time horizon of interest; 2) spatial constraint that guarantees the total power of all appliance at any time being smaller than the physical capacity of power line. There has been a mass of research works considering temporal and spatial constraints in their models [2], [7], [11], [14], [18], which means problems with these constraints are very important. From the optimization perspective, temporal constraint of one appliance couples its power scheduling decisions at different time slots, while spatial constraint couples the decisions of one appliance with those of other appliances. Enhancing computational efficiency is always of great importance in algorithm design, which enables faster response in real time computation or cheaper hardware cost in algorithm implementation. To this end, this paper proposes a faster algorithm by fully exploiting the structural advantages of the power scheduling problem with spatially and temporally coupled constraints.

For the ease of understanding, we first propose a fast algorithm based on KKT conditions to solve the optimal scheduling problem of a single appliance with temporally coupled constraint. The efficiency of this method origins from transferring a multi-variable optimization problem into a single-variable one. Then, we further extend the proposed method to the multi-appliance case, which is subject to both temporally and spatially coupled constraints. Besides the computation efficiency, the proposed method also offers strong robustness against the variation of parameter settings. The performance of many optimization algorithms heavily rely on the choice of step size: a large step size enjoys computational speed but suffers from the risk of non-convergence due to different parameter settings of the problem; and vice versa for a small step size. Being different from these approaches, the proposed algorithm does not have a step size. Therefore, it is much less sensitive to the parameter setting of the problem, and always keeps high success rate in solving scheduling problems.

The rest of the paper is organized as follows: Section II establishes the model for scheduling single appliance and proposes our algorithm. Section III extends the proposed algorithm to multi-appliance scheduling problem. Section IV briefly revisits dual decomposition method as a baseline for performance comparison. Section V shows the numerical results of our algorithm compared with dual decomposition and interior point method. Conclusions are drawn in Section VI.

II. Fast Algorithm for Single Appliance Scheduling With Temporally Coupled Constraint

In this section, we first establish the model for single appliance scheduling problem with temporally coupled constraint. Then we propose an algorithm based KKT conditions to fast solve the optimization problem.

A. Single Appliance Scheduling Model

We first establish the model for power scheduling of a single appliance. Let $\mathcal{H} \triangleq [1, \ldots, H]$ denote the workable time slots for a shiftable appliance. Let T be the duration of a slot, and without loss of generality assume $T = 1$. $P \triangleq [p^1, \ldots, p^H]$ is the power vector of the appliance, where p^h is the power of the appliance at time slot h. Due to the time and power limits, p^h is constrained as

$$p^{min} \leq p^h \leq p^{max}, \;\; h \in \mathcal{H} \tag{1}$$

where p^{min} and p^{max} are the minimum and maximum power limits of the appliance, respectively. The accumulative energy consumption of the appliance in all slots should exceed a given amount D to complete a task, i.e.,

$$T \sum_{h \in \mathcal{H}} p^h = \sum_{h \in \mathcal{H}} p^h \geq D \tag{2}$$

which is a temporal constraint coupling p^h in all slots.

When no scheduling is applied, the appliance is operated at the constant rated power within the power limit:

$$\overline{p} = D/H \in \left(p^{min}, p^{max}\right) \tag{3}$$

When the appliance is scheduled, p^h may deviate from $\overline{p}$ and lead to a certain cost, e.g., the dissatisfaction of users, the wear and tear of the appliance, battery degradation, etc. In general, this cost can be formulated as $W(\Delta p^h)$, where $\Delta p^h = p^h - \overline{p}$ is the deviation of actual power. The approach proposed in this paper is general to any form of cost function, as long as $W(\Delta p^h)$ is a strictly convex function with $\Delta p^h = 0$ being the minimum point and $W(0) = 0$. For example, the quadratic function $W(\Delta p^h) = (p^h - \overline{p})^2$ [19], [20]. In fact, almost all the existing works about optimal scheduling problem assume that the cost functions are quadratic or convex [14], [16], [17], [19], [21], [22], which is reasonable and general enough for most cases. Thus we design algorithms based for convex cost function, and non-convex case is out of our scope.

From above, we establish the optimization problem for single appliance:

$$\begin{aligned} \min_{p^h} \ & \sum_{h\in\mathcal{H}} \Big[\alpha^h p^h + oW\Big(\Delta p^h\Big)\Big] \\ s.t. \ & D - \sum_{h\in\mathcal{H}} p^h \le 0 \\ & p^{min} \le p^h \le p^{max}, \quad \forall h \in \mathcal{H} \end{aligned} \tag{4}$$

where α^h is the electricity price at slot h, o is the weighting factor.

Our model considers power scheduling of appliances with temporal constraint and spatial constraint in a certain time period of interest. There has been many studies about demand response management employ similar models [2], [10], [17], [21], [23]. Although the model is simple, under sound assumptions and modifications, it can still be adopted to handle other complicated appliances, such as batteries, electric vehicles (EVs) and water heaters, whose scheduling involves inter-temporal constraints, besides the temporal and spatial constraints in (4). However, if these inter-temporal constraints are always satisfied, which is often the case in a short period of time, our approach is still applicable. These cases appear when, for example, batteries are scheduled to discharge to meet an outage emergency, or EVs are scheduled to charge to a certain level, or the water need to be heated from a certain temperature to a desired one.

Furthermore, our approach can be embedded into other optimization structures as a core algorithm for convex power scheduling problems with high efficiency. For example, it can be embedded into branch and bound method to handle the cases where discrete or binary variables are present, into sliding time window structure to address uncertainty existing in the supply and demand side, and into two-stage settlement model to handle more practical scenarios.

B. Fast Algorithm Based on KKT Conditions

Clearly, (4) is a convex problem, and can be solved by conventional methods, e.g., interior points method. But the computational speed cannot meet the requirement of real-time DRM for a large number of appliances, thus a more efficient method is designed based on KKT conditions [24] as below

$$\begin{cases} \alpha^h + oW'\big(\Delta p^h\big) - \tau + \psi^h - \xi^h = 0 \\ 0 \le \tau \perp \big(\sum_{h\in H} p^h - D\big) \ge 0 \\ 0 \le \psi^h \perp \big(p^{max} - p^h\big) \ge 0 \\ 0 \le \xi^h \perp \big(p^h - p^{min}\big) \ge 0 \end{cases}$$

where τ, ψ^h and ξ^h are KKT operators associated with corresponding constraints, and W' is the derivative function of W. We can take τ as a variable to solve above problem and obtain the analytical solution of power p^h,

$$p^h(\tau) = \max\left\{\min\left\{W'^{-1}\left(\frac{\tau - \alpha^h}{o}\right) + \overline{p}, p^{max}\right\}, p^{min}\right\} \tag{5}$$

where W'^{-1} is the inverse function of W'. The derivation of (5) is detailed in following proof.

Proof: From the first order optimal condition, we have $\alpha^h + oW'(\Delta p^h) - \tau + \psi^h - \xi^h = 0$, thus $p^h = W'^{-1}(\frac{\tau - \psi^h + \xi^h - \alpha^h}{o}) + \overline{p}$. If $p^{min} < p^h < p^{max}$, we have $\psi^h = \xi^h = 0$ based on complementary conditions, thus $p^{min} < p^h = W'^{-1}(\frac{\tau-\alpha^h}{o}) + \overline{p} < p^{max}$. If $p^h = p^{max}$, we have $\psi^h > 0$ and $\xi^h = 0$, thus $p^{min} < p^h = W'^{-1}(\frac{\tau-\psi^h-\alpha^h}{o}) + \overline{p} = p^{max} < W'^{-1}(\frac{\tau-\alpha^h}{o}) + \overline{p}$. Similarly, if $p^h = p^{min}$, $W'^{-1}(\frac{\tau-\alpha^h}{o}) + \overline{p} < p^h = W'^{-1}(\frac{\tau+\xi^h-\alpha^h}{o}) + \overline{p} = p^{min} < p^{max}$. Therefore, the equation $p^h(\tau) = \max\{\min\{W'^{-1}(\frac{\tau-\alpha^h}{o}) + \overline{p}, p^{max}\}, p^{min}\}$ can cover above all three cases into one analytical formulation. ■

Obviously, (5) indicates that once the optimal τ^* is determined, the optimal p^h is also determined. Thus, the original H-dimensional optimization problem has been converted into an 1-dimensional problem.

To solve τ^*, it is noticed that $\tau(D - \sum p^h) = 0$, which implies two possibilities: 1) $\tau^* = 0$ or 2)

$$\tilde{D}\big(\tau^*\big) \triangleq \sum_{h\in\mathcal{H}} p^h\big(\tau^*\big) = D \tag{6}$$

A close examination can exclude the possibility of case 1) as follows. $p^h(\tau)$ is a piecewise and monotonically increasing function of τ, and $\tilde{D}(\tau)$ has the same properties because $\tilde{D}(\tau)$ is the sum of all $p^h(\tau)$. Thus $\tilde{D}(\tau) \ge \tilde{D}(0)$ for $\tau \ge 0$. If $\tau^* = 0$, the corresponding power is

$$p^h(0) = \max\left\{\min\left\{W'^{-1}\left(\frac{-\alpha^h}{o}\right) + \overline{p}, p^{max}\right\}, p^{min}\right\}$$

More specifically, because $W'^{-1}(\frac{-\alpha^h}{o})$ is negative and $\overline{p} \in (p^{min}, p^{max})$, the value of $p^h(0)$ can only be $W'^{-1}(\frac{-\alpha^h}{o}) + \overline{p}$ or p^{min}, and both values are smaller than $\overline{p}$. Thus $\tilde{D}(0) < \sum \overline{p} = D$, and only the second case (6) is possible.

Because of the piecewise and monotonically increasing properties of $\tilde{D}(\tau)$, a simple yet efficient method is proposed to solve the scheduling problem. The key idea is presented below.

As shown in Fig. 1(a), $p^h(\tau)$ is a piecewise function of τ with two non-differentiable breaking points

$$\begin{cases} \tau_1 = oW'\big(\Delta p^{min}\big) + \alpha^h \\ \tau_2 = oW'(\Delta p^{max}) + \alpha^h \end{cases} \tag{7}$$

Thus, $p^h(\tau)$ consists of three segments, i.e.,

$$p^h = \begin{cases} p^{min}, & 0 \le \tau \le \tau_1 \\ W'^{-1}\Big(\frac{\tau-\alpha^h}{o}\Big) + \overline{p}, & \tau_1 \le \tau \le \tau_2 \\ p^{max}, & \tau \ge \tau_2 \end{cases} \tag{8}$$

In each segment, the profile is continuous and differentiable everywhere. Because $\tilde{D}(\tau)$ is the sum of $p^h(\tau)$, it has $2H$

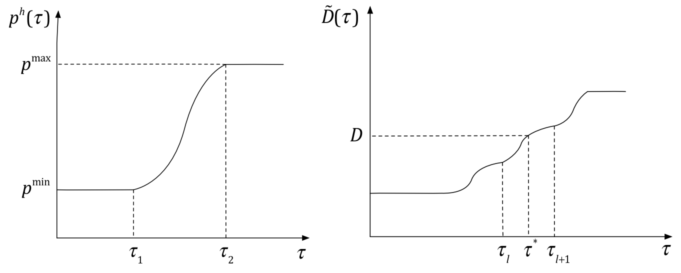

Fig. 1. Profiles of $p^h(\tau_a)$ and $\tilde{D}(\tau)$.

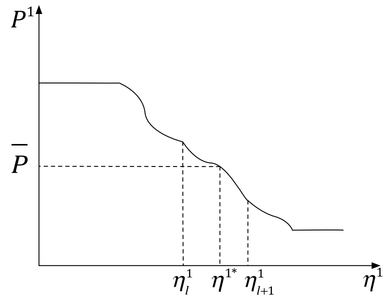

Fig. 2. Profile of P^1.

breaking points at most, and its shape is shown in Fig. 1 (b). Meanwhile, it is notable that all these $2H$ breaking points are non-differentiable but can be readily obtained from (7). Fig. 1(b) also demonstrates how to solve (6): first, locate two breaking points τ_l and τ_{l+1} that contain τ^* in between; then, utilize the gradient information to quickly compute τ^*.

As detailed in Algorithm 1, in the first step we shall reorder $2H$ breaking points in ascending order and find the pair τ_l and τ_{l+1} via binary search, such that $\tilde{D}(\tau_l) \leq D \leq \tilde{D}(\tau_{l+1})$. Thus τ^* lies between τ_l and τ_{l+1}. Second, iteratively approach τ^* via Newton downhill method [25] with the gradient of this segment as below

$$\tau_{n+1}^* = \tau_n^* - \lambda_n \frac{\tilde{D}\left(\tau_n^*\right) - D}{\tilde{D}'\left(\tau_n^*\right)} \tag{9}$$

where $n \in \mathbb{N}^+$ denotes the index of iterations, and λ_n is initialized to be 1 at each iteration. If $|\tilde{D}(\tau_{n+1}^*) - D| < |\tilde{D}(\tau_n^*) - D|$, $\lambda_n = 1$; if not, change $\lambda_n = 0.5 * \lambda_n$ until the inequality satisfied. $\tilde{D}'(\tau)$ is the derivative of accumulative energy consumption of the appliance in all slots to τ as below

$$\frac{\mathrm{d}\tilde{D}(\tau)}{\mathrm{d}\tau} = \frac{1}{o} \sum_{h \in \mathcal{H}} S\left(p^h\right) \tag{10}$$

where $S(p^h)$ is the derivative function of $W'^{-1}(x)$ at $x = \frac{\tau - \alpha^h}{o}$. The derivation of (10) is given in Appendix A. Finally, obtain $p^h(\tau^*)$ according to (5).

In summary, the proposed method converts solving original problem in H-dimension space into solving τ^* in 1-dimension space. Thus, the computational speed can be greatly improved.

Turning to the convergence rate, the algorithm consists of bi-section method and Newton method. Bi-section method is linear convergence, while Newton method is quadratic convergence [26].

Algorithm 1 Single Appliance Scheduling

Step 1: Bisection Search
Initialize $l = 1, r = 2H$.
while $r - l > 1$ **do**
  $m = [(l + r)/2]$ and $\tilde{D} = \sum_{h \in H} p^h(\tau_m)$.
  if $\tilde{D} = D$ **then**
    $\tau^* = \tau_m$.
  else if $\tilde{D} < D$ **then**
    $l = m$
  else
    $r = m$
  end if
end while
Step 2: Newton Downhill Method
Obtain the initial guess τ_l and initial value $\tilde{D}(\tau_l) = \sum_{h \in H} p^h(\tau_l)$.
while $\left|\tilde{D}(\tau_l) - D\right| > \varepsilon$ **do**
  Set $\lambda_l = 1$, update $\tau_{l+1} = \tau_l - \lambda_l \frac{\tilde{D}(\tau_l) - D}{\tilde{D}'(\tau_l)}$
  obtain $\tilde{D}(\tau_{l+1})$ and $\tilde{D}'(\tau_{l+1})$.
  while $\left|\tilde{D}(\tau_{l+1}) - D\right| \geqslant \left|\tilde{D}(\tau_l) - D\right|$ **do**
    λ_l=0.5*λ_l, update $\tau_{l+1} = \tau_l - \lambda_l \frac{\tilde{D}(\tau_l) - D}{\tilde{D}'(\tau_l)}$;
    obtain $\tilde{D}(\tau_{l+1})$ and $\tilde{D}'(\tau_{l+1})$.
  end while
  $l = l + 1$.
end while
$p^h = p^h(\tau^*)$

III. Extension to Multiple Appliances With Temporally-Spatially Coupled Constraints

Next, we extend this method to efficiently solve the power scheduling problem of a large number of appliances. Let $\mathcal{A}$ denote a set of shiftable appliances, and A be the cardinal number of set $\mathcal{A}$. In addition to temporally coupled constraints, there exists spatially coupled constraints in each slot, which implies the total power of all appliances bounded by the physical power limit of the district transformer or substation transformer $\overline{P}$, as below:

$$\sum_{a \in \mathcal{A}} p_a^h \leq \overline{P} \in \left(\sum_{a \in \mathcal{A}} \overline{p}_a, \sum_{a \in \mathcal{A}} p_a^{max} \right), \forall h \in \mathcal{H} \tag{11}$$

Note that imposing the lower bound on $\overline{P}$ ensures the feasibility of the problem, while the upper bound ensures the line limit $\overline{P}$ being activated at certain slots. Then, the optimization problem can be established for multiple appliances:

$$\begin{aligned} \min_{p_a^h} \quad & \sum_{h \in \mathcal{H}} \sum_{a \in \mathcal{A}} \left[\alpha^h p_a^h + o_a W_a\left(\Delta p_a^h\right) \right] \\ s.t. \quad & D_a - \sum_{h \in \mathcal{H}} p_a^h \leq 0, \quad \forall a \in \mathcal{A} \\ & \sum_{a \in \mathcal{A}} p_a^h - \overline{P} \leq 0, \quad \forall h \in \mathcal{H} \\ & p_a^{min} \leq p_a^h \leq p_a^{max}, \quad \forall a \in \mathcal{A}, \ \forall h \in \mathcal{H} \end{aligned} \tag{12}$$

Note that the first two constraints are temporal and spatial constraints that couple p_a^h in all time slots and all appliances, respectively. Consequently, the KKT conditions can be written as:

$$\begin{cases} \alpha^h + o_a W'_a\left(\Delta p_a^h\right) - \tau_a + \eta^h + \psi_a^h - \xi_a^h = 0 \\ 0 \le \tau_a \perp \left(\sum_{h\in\mathcal{H}} p_a^h - D_a\right) \ge 0 \\ 0 \le \eta^h \perp \left(\overline{P} - \sum_{a\in\mathcal{A}} p_a^h\right) \ge 0 \\ 0 \le \psi_a^h \perp \left(p_a^{max} - p_a^h\right) \ge 0 \\ 0 \le \xi_a^h \perp \left(p_a^h - p_a^{min}\right) \ge 0 \end{cases}$$

where τ_a, η^h, ψ_a^h and ξ_a^h are KKT operators associated with corresponding constraints. Similar to (5), we take τ_a and η^h as variables to solve the equations and obtain the analytical representation of power p_a^h

$$p_a^h\left(\tau_a, \eta^h\right) = \max\left\{\min\left\{W'^{-1}_a\left(\frac{\tau_a - \alpha^h - \eta^h}{o_a}\right) + \overline{p}_a, p_a^{max}\right\}, p_a^{min}\right\} \tag{13}$$

It is clear from (13) that instead of solving all $A \times H$ variables p_a^h, we only need to find $A + H$ operators τ_a and η^h, and obviously $A + H$ is much smaller than $A \times H$ in a large scale problem.

Then, we shall show that $\eta^h, h = 1, 2, \ldots, H$ are correlated as stated later in *Lemma 1*. Before approaching this result and for the ease of presentation, we first reorder $\mathcal{H}$ in price ascending order, i.e., $\alpha^1 \le \alpha^2 \le \cdots \le \alpha^H$. Undoubtedly, the total power of all appliances is higher at low-price slots, and lower at high-price slots. Assume total power of all appliances reaches the upper limit $\overline{P}$ at the first k time slots

$$\sum_{a\in\mathcal{A}} p_a^1\left(\tau_a, \eta^1\right) = \sum_{a\in\mathcal{A}} p_a^2\left(\tau_a, \eta^2\right) = \cdots = \sum_{a\in\mathcal{A}} p_a^k\left(\tau_a, \eta^k\right) = \overline{P} \tag{14}$$

which are referred to as the "full power slots" in later discussion. Then, we have the following lemma.

Lemma 1: $\alpha^1 + \eta^1 = \alpha^2 + \eta^2 = \cdots = \alpha^k + \eta^k$.

Proof: We prove *Lemma 1* by contradiction. If *Lemma 1* is false, there must exist a minimum value and we assume it is $\alpha^{h_1} + \eta^{h_1}$. Because $\overline{P} < \sum_{a\in\mathcal{A}} p_a^{max}$, at least one appliance's power is below its maximum at slot h_1. Then at slot h_1, for those appliances whose power is below its maximum, their power at slot h_1 is larger than the power at other slots from (13) because of the monotonically increasing property of W'^{-1}_a. It results in $\sum_{a\in\mathcal{A}} p_a^{h_1}(\tau_a) > \sum_{a\in\mathcal{A}} p_a^h(\tau_a)$ for $h \ne h_1$, which contradicts (14). Hence, *Lemma 1* holds. ■

On the other hand, except for these k slots, the total power of all appliances at other slots is smaller than $\overline{P}$, thus the corresponding KKT operator η^h is equal to zero. Consequently, we have

$$\eta^h = \max\left\{\eta^1 + \alpha^1 - \alpha^h, 0\right\}, \forall h \in \mathcal{H} \tag{15}$$

From (15) it is clear that once η^1 is yielded, all other η^h can be readily obtained. Therefore, we shall present the method of computing η^1 next.

The key idea of finding η^1 via iterations can be summarized below. For an initial guess of η^1, other η^h can be directly obtained from (15). Then, by ignoring the total power constraint, problem (12) retreats to problem (4). Thus, each appliance can compute τ_a and p_a^1 separately, with known $\eta^h, h = 1, 2, \ldots, H$. Then we can obtain the total power at first slot $P^1 \triangleq \sum_{a\in\mathcal{A}} p_a^1(\tau_a, \eta^1)$, as well as the difference with $\overline{P}$. Consequently, η^1 will be updated iteratively until $P^1 = \overline{P}$. For efficiently updating, the key properties of P^1 with respect to η^1 are analyzed below.

Property 1: Except for finite breaking points, P^1 is differentiable with respect to η^1, and $\partial P^1/\partial\eta^1 = -\sum_{a\in\mathcal{A}'} [S_a(p_a^1) \sum_{h=k+1}^H S_a(p_a^h)]/[o_a \sum_{h=1}^H S_a(p_a^h)]$, where $\mathcal{A}'$ is the set of appliances whose power are below the maximum p_a^{max} given a η^1, and $S_a(p_a^h)$ is the derivative function of $W'^{-1}_a(x)$ at $x = \frac{\tau_a - \alpha^h - \eta^h}{o_a}$.

From *Property 1*, since $S_a(p_a^h) > 0$, P^1 is monotonically decreasing in η^1 with finite breaking points as shown in Fig. 2. Intuitively, when η^1 is increasingly passing through certain values, the sets $\mathcal{A}'$ and $[k+1, H]$ will change, which makes $\partial P^1/\partial\eta^1$ piecewise differentiable. These values are referred to as breaking points. Consequently, we show the property of breaking points.

Property 2: There are two types of breaking points: 1) $\alpha^k - \alpha^1, k = 1, 2, \ldots, H$; 2) $\max\{\tau_a(0) - \alpha^1 - o_a W'_a(p_a^{max} - \overline{p}_a), 0\}, \forall a \in \mathcal{A}$. These breaking points are non-differentiable.

In details, when passing through the first type of breaking points, k will increase, and $|\mathcal{A}'|$ increases when passing through the second type of breaking points. The derivation and more details are given in Appendix B and C.

We define $\mathcal{A}'(0)$ as the set of appliances whose power are below the maximum in the first slot when $\eta^1 = 0$, then we have $W'^{-1}_{a_1}(\frac{\tau_{a_1}(0)-\alpha^h}{o_{a_1}}) + \overline{p}_{a_1} < p_{a_1}^{max}$, i.e., $W'^{-1}_{a_1}(\frac{\tau_{a_1}(0)-\alpha^h}{o_{a_1}}) < p_{a_1}^{max} - \overline{p}_{a_1}$. Since inverse function does not change the monotonicity of primitive, we have $\frac{\tau_{a_1}(0)-\alpha^h}{o_{a_1}} < W'_{a_1}(p_{a_1}^{max} - \overline{p}_{a_1})$. Hence, the equation $\eta_{a_1}^1 = \tau_{a_1}(0) - \alpha^1 - o_{a_1} W'_{a_1}(p_{a_1}^{max} - \overline{p}_{a_1}) < \tau_{a_1}(0) - \alpha^1 - o_{a_1}\frac{\tau_{a_1}(0)-\alpha^h}{o_{a_1}} = 0$, which means the value of the second type breaking points of these appliances are zero. Thus, there are at most $H + A - |\mathcal{A}'(0)|$ positive breaking points, reorder them in an ascending order and construct the sequence of breaking points, which is $\boldsymbol{\eta^1} \triangleq [\eta_1^1, \eta_2^1, \ldots, \eta_{H+A-|\mathcal{A}'(0)|}^1]$. Similar to the single appliance case, binary search can be applied to find a pair of η^1, e.g., η_l^1 and η_{l+1}^1, such that $\sum_{a\in\mathcal{A}} p_a^1(\eta_l^1) \le \overline{P} \le \sum_{a\in\mathcal{A}} p_a^1(\eta_{l+1}^1)$. Then, η^{1*} such that $P^1 = \overline{P}$ can be obtained via Newton downhill method with known gradient of this segment according to *Property 3*. It is notable that both binary search and Newton downhill method can be performed in a distributed manner: every time the central controller announces a testing η_m^1, then each appliance feedbacks its p_a^1 and S_a for updating. The details are summarized in Algorithm 2.

IV. Revisit of Two-Level Dual Decomposition Method

To compare the performance of the proposed method, this section revisits the dual decomposition method which

Algorithm 2 Multiple Appliances Scheduling

Step 1: Information collection
The central controller broadcasts $\eta^1 = 0$ to the controller of each appliance.
Each appliance returns p_a^1, o_a and the second type breaking points η^1.
Obtains $P^1 = \sum_{a\in A} p_a^1$ and $\boldsymbol{\eta}^1$. Set $l = 1, r = H + A - |\mathcal{A}'(0)|$ and $m = 1$.
Step 2: Bisection Search
if $P^1 \leq \overline{P}$ **then**
    $\eta^{1*} = 0$
else
    while $r - l > 1$ **do**
        $m = [(l + r)/2]$
        Broadcasts η_m^1;
        Each appliance returns p_a^1 to obtain P_m^1;
        if $P_m^1 = \overline{P}$ **then**
            $\eta^{1*} = \eta_m^1$
        else if $P_m^1 < \overline{P}$ **then**
            $r = m$
        else
            $l = m$
        end if
    end while
end if
Step 3: Newton Downhill Method
Broadcasts η_l^1 and set $n = 1$
Each appliance returns p_a^1 and $S_a(p_a^h)$
Obtains P_n^1 and $P_n^{1'}$;
while $\left|P_n^1 - \overline{P}\right| > \varepsilon$ **do**
    Set $\lambda_n = 1$, update $\eta_{n+1}^1 = \eta_n^1 - \lambda_n \frac{P_n^1 - \overline{P}}{P_n^{1'}}$, broadcast η_{n+1}^1;
    Each appliance returns p_a^1 and $S_a(p_a^h)$;
    Obtains P_{n+1}^1 and $P_{n+1}^{1'}$.
    while $\left|P_{n+1}^1 - \overline{P}\right| \geqslant \left|P_n^1 - \overline{P}\right|$ **do**
        λ_n=0.5*λ_n, $\eta_{n+1}^1 = \eta_n^1 - \lambda_n \frac{P_n^1 - \overline{P}}{P_n^{1'}}$, broadcast η_{n+1}^1;
        Each appliance returns p_a^1 and $S_a(p_a^h)$;
        Obtains P_{n+1}^1 and $P_{n+1}^{1'}$.
    end while
    $n = n + 1$
end while
$p_a^h = p_a^h(\tau^a, \eta^{1*})$

is widely used to solve the optimal power scheduling problem [7], [8], [14]. Two-level dual decomposition method is proposed to decouple the problem spatially and temporally, respectively. The first level is applied to decouple the problem spatially, which allows the appliances to solve their subproblems individually. Using the KKT operator η^h, the Lagrangian of multi-appliance problem is

$$L\left(p_a^h, \eta^h\right) = \sum_{h\in\mathcal{H}} \sum_{a\in\mathcal{A}} \left[\alpha^h p_a^h + o_a W_a\left(\Delta p_a^h\right)\right] + \eta^h \left(\sum_{a\in\mathcal{A}} p_a^h - \overline{P}\right)$$

The primal problem is equivalent to the following equation

$$\mathbb{P}: \min_{p_a^h} \max_{\eta^h} L\left(p_a^h, \eta^h\right) \tag{16}$$

while the dual problem is

$$\mathbb{D}: \max_{\eta^h} \min_{p_a^h} L\left(p_a^h, \eta^h\right) = \max_{\eta^h} \sum_{a\in\mathcal{A}} L_a + \max_{\eta^h}\left(-\eta^h \overline{P}\right) \tag{17}$$

The outer layer is to maximize the dual function over η^h, while the inner layer is the subproblem of each appliance:

$$L_a = \min_{p_a^h} \sum_{h\in\mathcal{H}} \left(\alpha^h + \eta^h\right) p_a^h + o_a W_a\left(\Delta p_a^h\right) \tag{18}$$

Then for the subproblem of each appliance, the second level is applied to decouple the subproblem temporally into single-slot problem. Using the KKT operator τ_a, the Lagrangian of the subproblem is

$$L_a\left(p_a^h, \tau_a\right) = \sum_{h\in\mathcal{H}} \left[\left(\alpha^h + \eta^h\right) p_a^h + o_a W_a\left(\Delta p_a^h\right)\right] + \tau_a \left(D_a - \sum_{h\in\mathcal{A}} p_a^h\right)$$

The primal problem is equivalent to the following equation

$$\mathbb{P}_a: \min_{p_a^h} \max_{\tau_a} L_a\left(p_a^h, \tau_a\right) \tag{19}$$

While the dual problem is

$$\mathbb{D}_a: \max_{\tau_a} \min_{p_a^h} L_a\left(p_a^h, \tau_a\right) = \max_{\tau_a} \sum_{h\in\mathcal{H}} L_a^h + \max_{\tau_a}(\tau_a D_a) \tag{20}$$

The outer layer is to maximize the dual function over τ_a, while the inner layer is the scheduling of power at each time slot, which can be solved easily from (13).

$$L_a^h = \min_{p_a^h}\left(\alpha^h + \eta^h - \tau_a\right) p_a^h + o_a W_a\left(\Delta p_a^h\right) \tag{21}$$

As the primal problem is convex but non-differentiable, the subgradient method which guarantees the optimality is introduced. The updating rules of η^h and τ_a are:

$$\begin{cases} \eta^h(k_1 + 1) = \left[\eta^h(k_1) - \gamma_\eta g^h(k_1)\right]^+ \\ \tau_a(k_2 + 1) = [\tau_a(k_2) - \delta_\tau h_a(k_2)]^+ \end{cases} \tag{22}$$

where k_1, k_2 and γ_η, δ_τ are the iteration indexes and step sizes, respectively. In order to guarantee convergence under different problem scales, the step sizes should be quite small but at the cost of computational speed [24]. $g^h(k_1)$ and $h_a(k_2)$ are the subgradients of the dual function with respect to γ_η and δ_τ, which are

$$\begin{cases} g^h(k_1) = -\sum_{a\in\mathcal{A}} p_a^h\left(\eta^h(k_1)\right) + \overline{P} \\ h_a(k_2) = \sum_{h\in\mathcal{H}} p_a^h(\tau_a(k_2)) - D_a \end{cases} \tag{23}$$

The two-level dual decomposition algorithm is elaborated as Algorithm 3.

Algorithm 3 Two-Level Dual Decomposition

Initialize iteration indexes $k_1 = 1, k_2 = 1$ and $\eta^h(k_1) = 0$, $\tau_a(k_2) = 0$.
repeat
  The central controller sends $\eta^h(k_1)$ to the controller of each appliance;
  repeat
    The appliance controller calculates p_a^h with $\eta^h(k_1)$ and $\tau_a(k_2)$;
    Controller of appliance updates $\tau_a(k_2 + 1)$
  until k_2 exceeds the maximum iteration number or minimum $h_a(k_2)$ is small enough.
  The central controller updates $\eta^h(k_1 + 1)$
until k_1 exceeds the maximum iteration number or minimum $g^h(k_1)$ is small enough.

V. Numerical Results

Numerical results are presented to compare the performance of our proposed algorithm with dual decomposition and interior point method. The setup is as follows. For the sake of simplicity, let H and A represent the number of time slots and appliances. To better compare the performance, uniform distribution stochastic parameter settings are applied. The electricity prices α^h varies from 1.5 to 3.5. The minimum power p_a^{min} and maximum power p_a^{max} of appliances vary from 0 to 1 and $p_a^{min}+4$ to $p_a^{min}+5$, respectively. The task to be finished D_a are $H \times p_a^{max} \times c$, and c is a coefficient that varies from 0.3 to 0.5. The weighting factors o_a vary from 0.05 to 0.15. The total power constraint $\overline{P}$ varies from $2.5A$ to $3.5A$. Simulations have been performed 50 times for our algorithm, dual decomposition and interior point method at different scale of appliances by using MATLAB R2014b on a PC with 1.6 GHz Intel Core 2 Duo CPU and 4 GB memory.

We first investigate the single-appliance case. The cost function is often set to be a quadratic form as: $o(p^h - \overline{p})^2$[19]. Real time-of-use (TOU) price of 24 hours depicted in Fig. 3(a) is utilized in the simulation. Fig. 3(b) depicts the rated power without scheduling. Fig. 3(c)-3(d) are the scheduled consumption with two different weighing factors ($o = 0.2$ and 0.4). The result indicates that appliance can shift consumption off peak hours to save electricity bill but at the cost of deviation from rated power. It also demonstrates that appliance can shift more consumption when weighing factor is smaller because users attach more importance to electricity bill.

Then, we investigate the multi-appliance case. The performance is compared mainly in two aspects. The first is the computation speed under different number of appliances measuring by computation time. Since MATLAB cannot realize parallel computation, the computation times are obtained by taking the average time of serial times for both our method and dual decomposition method, which guarantees the fairness of results. Moreover, the computational times of single appliance scheduling from run to run are quite close, thus the average time is a good approximation. The second is the sensitivity to parameter setting measuring by variance. In the simulation, we fix H to be 288, i.e., 5 minutes each slot, to demonstrate

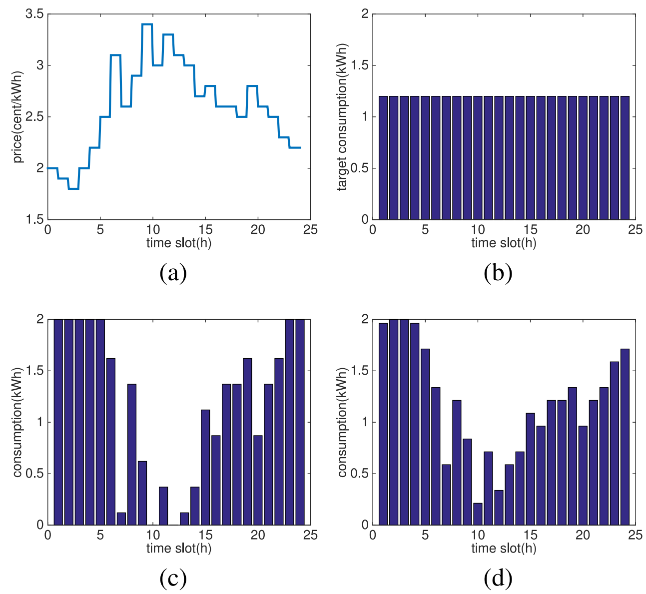

Fig. 3. Single-appliance case with different trade-off factors.

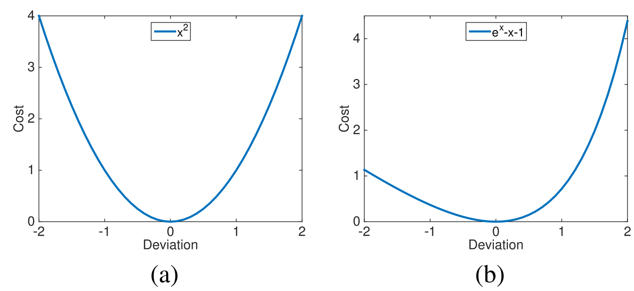

Fig. 4. Two different cost functions.

the performance with the increase of A. Two different cost functions are employed in our simulation, as shown in Fig. 4, a quadratic function $(p^h - \overline{p})^2$ and $e^{(p^h - \overline{p})} - (p^h - \overline{p}) - 1$. In dual decomposition method, the step sizes are fixed to be $\gamma_\eta = 0.0018, \delta_\tau = 0.001$ for the former function and $\gamma_\eta = 0.003, \delta_\tau = 0.0005$ for the latter, which are tested to be a suitable combination. In addition, we fix the maximum iteration number to be 500.

Fig. 5-6 indicate the performance of our algorithm, dual decomposition method and interior point method with two different convex cost functions. When it comes to our algorithm, the computation speed is obviously faster than other two methods with the same A in most cases and the computation time is almost kept constant. It also demonstrates that our algorithm has extreme low variance in computation time, which means low sensitivity to change of parameter setting. As for dual decomposition method, the iteration times are highly affected by parameter setting resulting in high variance as shown in Fig. 5. Its computation time almost keeps decreasing in the beginning before about 150 and 130 appliances for the two functions, respectively. It is because that more iterations are taken to converge at first due to small step size, then the iterations drop at a stable rate with the increase of appliance scale. However, after A increases over 150 and 130, the

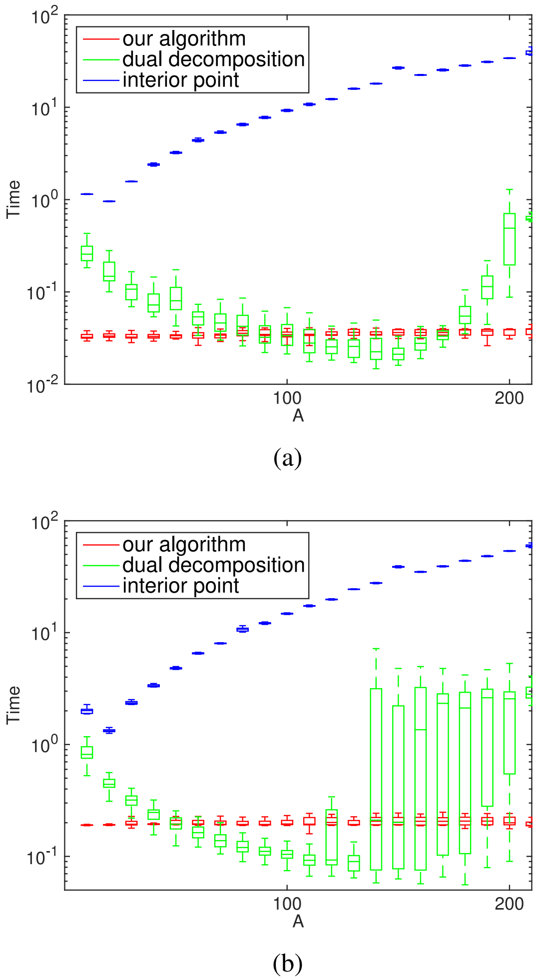

Fig. 5. Computation time of two functions.

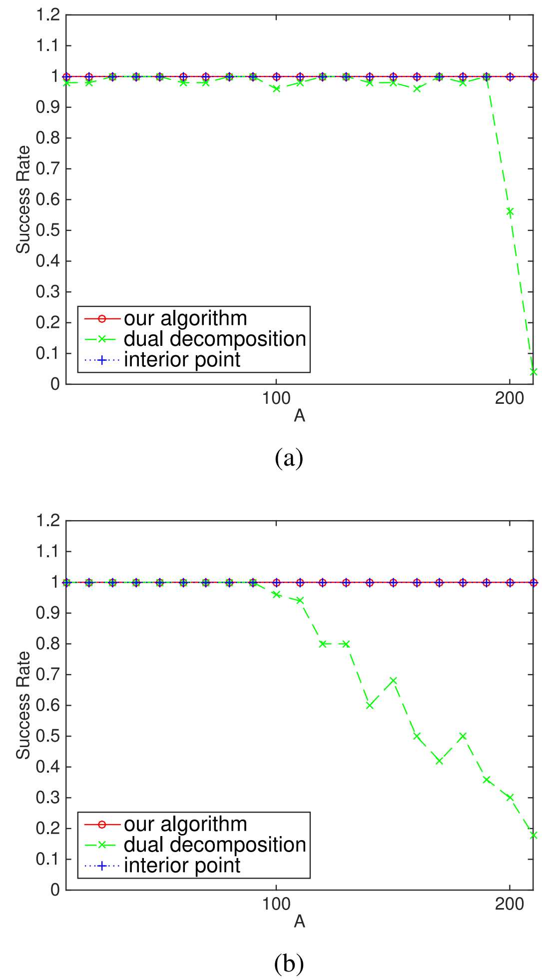

Fig. 6. Success rate of two functions.

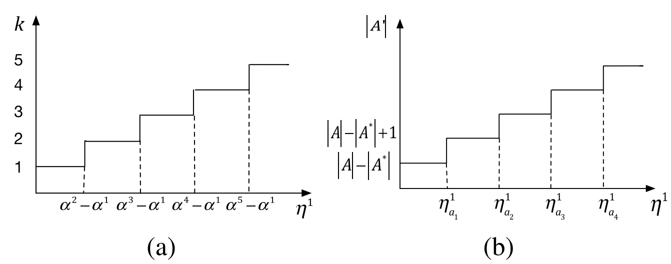

Fig. 7. Breaking points.

computation time and variance increase dramatically simultaneously. Turning to interior point method, the computation time is nearly exponential increasing with low variance.

Compared with interior point methods, there are several major differences as follows. First, our method is an exact way to calculate the optimal solution, and we do not approximate the problem into another one or modify the KKT conditions. Second, interior point methods put the inequality constraints into objective to make the problem solvable. According to duality theory, essentially, it does not enhance the computational efficiency very much. On the contrary, our method converts the high-dimension problem into a low-dimension one by exploring the structure information between KKT operators, and solve the problem with much higher computational efficiency.

In terms of the success rate, as shown in Fig. 6, dual decomposition method begins decreasing when A reaches 190, and drops sharply to zero in the end for the first function. As for the other function, as shown in Fig. 6, it keeps full in the beginning before 90 appliances. Subsequently, the rate declines stably until to 0.2 at last. In conclusion, due to the fixed step sizes, the dual decomposition will oscillate around the optimal point even fail to converge when A reaches a certain value. While the other two methods are highly robust that keep full success rate for all cases.

VI. Conclusion

Appliances have the ability to cut down and shift power consumption in order to save cost and reduce dissatisfaction. We propose a fast algorithm based on KKT conditions to solve a single-appliance power scheduling problem under real time price, and then extend to multi-appliance case with temporally-spatially coupled constraints. Numerical results show our algorithm is more computationally efficient and less sensitive to parameter setting compared with dual decomposition and interior point method.

Appendix A
Derivation of (10)

For simplicity, let $S(p^h)$ be the derivative function of $W'^{-1}(x)$ at $x=\frac{\tau-\alpha^h}{o}$, i.e.,

$$S\left(p^h\right)=\lim_{x\to\frac{\tau-\alpha^h}{o}}\frac{W'^{-1}(x)-W'^{-1}\left(\frac{\tau-\alpha^h}{o}\right)}{x-\frac{\tau-\alpha^h}{o}} \tag{24}$$

where $W'^{-1}(\frac{\tau-\alpha^h}{o})=p^h-\overline{p}$. If $p^h=p^{max}$ or p^{min}, $S(p^h)=0$. When τ has an increment $\Delta\tau$, the power in slot h increases by $W'^{-1}(\frac{\tau+\Delta\tau-\alpha^h}{o})-W'^{-1}(\frac{\tau-\alpha^h}{o})$. Thus the gradient of power in slot h to τ is

$$\begin{aligned}\frac{\mathrm{d}p^h}{\mathrm{d}\tau}&=\lim_{\Delta\tau\to0}\frac{W'^{-1}\left(\frac{\tau+\Delta\tau-\alpha^h}{o}\right)-W'^{-1}\left(\frac{\tau-\alpha^h}{o}\right)}{\Delta\tau}\\&=\lim_{\Delta\tau\to0}\frac{W'^{-1}\left(\frac{\tau+\Delta\tau-\alpha^h}{o}\right)-W'^{-1}\left(\frac{\tau-\alpha^h}{o}\right)}{\frac{\tau+\Delta\tau-\alpha^h}{o}-\frac{\tau-\alpha^h}{o}}\times\frac{1}{o}\\&=\frac{S\left(p^h\right)}{o}\end{aligned} \tag{25}$$

Hence, for $\tilde{D}(\tau)\triangleq\sum_{h\in\mathcal{H}}p^h(\tau)$

$$\frac{\partial\tilde{D}(\tau)}{\partial\tau}=\frac{1}{o}\sum_{h\in\mathcal{H}}S\left(p^h\right) \tag{26}$$

From (26), it is evident that the breaking points are non-differentiable, that is the reason why we need to determine the pair of breaking points where τ^* lies firstly.

Appendix B
Proof of Property 1

For simplicity, let $S_a(p_a^h)$ be the derivative function of $W'^{-1}_a(x)$ at $x=\frac{\tau_a-\alpha^h-\eta^h}{o_a}$, i.e.,

$$S_a\left(p_a^h\right)=\lim_{x\to\frac{\tau_a-\alpha^h-\eta^h}{o_a}}\frac{W'^{-1}_a(x)-W'^{-1}_a\left(\frac{\tau_a-\alpha^h-\eta^h}{o_a}\right)}{x-\frac{\tau_a-\alpha^h-\eta^h}{o_a}} \tag{27}$$

where $W'^{-1}_a(\frac{\tau_a-\alpha^h-\eta^h}{o_a})=p_a^h-\overline{p}_a$. Then, for each appliance $a\in\mathcal{A}'$, assume η^1 has a small increment $\Delta\eta^1$. From (15) η^h will also increase $\Delta\eta^1$ for $h\le k$, and keep zero for $h>k$. From (13) τ_a will also have an increment, denoted by $\Delta\tau_a$, to fulfill temporal constraint. Thus at slot $h>k$, the power increases by $W'^{-1}_a(\frac{\tau_a+\Delta\tau_a-\alpha^h-\eta^h}{o_a})-(p_a^h-\overline{p}_a)$, which means the power at slot $h\le k$ decreases by $(p_a^1-\overline{p}_a)-W'^{-1}_a(\frac{\tau_a+\Delta\tau_a-\alpha^1-\eta^1-\Delta\eta^1}{o_a})$. In order to fulfill the total task temporal constraint, the accumulative decrement of power at the first k slots should be equal to the accumulative increment of power in the remaining slots as below

$$\begin{aligned}&k\left[(p_a^1-\overline{p}_a)-W'^{-1}_a\left(\frac{\tau_a-\alpha^1-\eta^1}{o_a}-\frac{\Delta\eta^1-\Delta\tau_a}{o_a}\right)\right]\\&\quad=\sum_{h=k+1}^{H}\left[W'^{-1}_a\left(\frac{\tau_a-\alpha^h-\eta^h}{o_a}+\frac{\Delta\tau_a}{o_a}\right)-\left(p_a^h-\overline{p}_a\right)\right]\end{aligned} \tag{28}$$

When $\Delta\eta^1\to0$, $\frac{\Delta\eta^1-\Delta\tau_a}{o_a}\to0$, thus we can obtain

$$\begin{aligned}&k\lim\frac{\left(p_a^1-\overline{p}_a\right)-W'^{-1}_a\left(\frac{\tau_a-\alpha^1-\eta^1}{o_a}-\frac{\Delta\eta^1-\Delta\tau_a}{o_a}\right)}{\frac{\Delta\eta^1-\Delta\tau_a}{o_a}}\\&\quad\times\frac{\Delta\eta^1-\Delta\tau_a}{o_a}\\&\quad=\sum_{h=k+1}^{H}\lim\frac{W'^{-1}_a\left(\frac{\tau_a-\alpha^h-\eta^h}{o_a}+\frac{\Delta\tau_a}{o_a}\right)-\left(p_a^h-\overline{p}_a\right)}{\frac{\Delta\tau_a}{o_a}}\\&\quad\times\frac{\Delta\tau_a}{o_a}\end{aligned} \tag{29}$$

which equals to

$$\begin{aligned}&kS_a\left(p_a^1\right)\times\frac{\Delta\eta^1-\Delta\tau_a}{o_a}=\sum_{h=k+1}^{H}S_a\left(p_a^h\right)\times\frac{\Delta\tau_a}{o_a}\\&\Rightarrow\left[kS_a\left(p_a^1\right)+\sum_{h=k+1}^{H}S_a\left(p_a^h\right)\right]\Delta\tau_a=kS_a\left(p_a^1\right)\Delta\eta^1\\&\Rightarrow\frac{\Delta\tau_a}{\Delta\eta^1}=\frac{kS_a\left(p_a^1\right)}{kS_a\left(p_a^1\right)+\sum_{h=k+1}^{H}S_a\left(p_a^h\right)}\end{aligned} \tag{30}$$

Then the accumulative variation of power in the first slot of all appliances $\Delta P^1=\sum_{a\in\mathcal{A}}p_a^1(\tau_a,\eta^1+\Delta\eta^1)-\sum_{a\in\mathcal{A}}p_a^1(\tau_a,\eta^1)$ is formulated as

$$\begin{aligned}&\Delta P^1\\&=\sum_{a\in\mathcal{A}'}\left[W'^{-1}_a\left(\frac{\tau_a-\alpha^h-\eta^h}{o_a}-\frac{\Delta\eta^1-\Delta\tau_a}{o_a}\right)-\left(p_a^1-\overline{p}_a\right)\right]\\&=\sum_{a\in\mathcal{A}'}\sum_{h=k+1}^{H}\frac{1}{k}\left[\left(p_a^h-\overline{p}_a\right)-W'^{-1}_a\left(\frac{\tau_a-\alpha^h-\eta^h}{o_a}+\frac{\Delta\tau_a}{o_a}\right)\right]\\&=-\frac{1}{k}\sum_{a\in\mathcal{A}'}\sum_{h=k+1}^{H}S_a\left(p_a^h\right)\times\frac{\Delta\tau_a}{o_a}\end{aligned} \tag{31}$$

Thus the derivative of P^1 to η^1 can be obtained as below

$$\begin{aligned}\frac{\partial P^1}{\partial\eta^1}&=-\frac{1}{k}\sum_{a\in\mathcal{A}'}\sum_{h=k+1}^{H}S_a\left(p_a^h\right)\times\frac{\Delta\tau_a}{o_a\Delta\eta^1}\\&=-\frac{1}{k}\sum_{a\in\mathcal{A}'}\sum_{h=k+1}^{H}S_a\left(p_a^h\right)\\&\quad\times\frac{kS_a\left(p_a^1\right)}{o_a\left[kS_a\left(p_a^1\right)+\sum_{h=k+1}^{H}S_a\left(p_a^h\right)\right]}\\&=-\sum_{a\in\mathcal{A}'}\frac{S_a\left(p_a^1\right)\sum_{h=k+1}^{H}S_a\left(p_a^h\right)}{o_a\left[kS_a\left(p_a^1\right)+\sum_{h=k+1}^{H}S_a\left(p_a^h\right)\right]}\\&=-\sum_{a\in\mathcal{A}'}\frac{S_a\left(p_a^1\right)\sum_{h=k+1}^{H}S_a\left(p_a^h\right)}{o_a\sum_{h=1}^{H}S_a\left(p_a^h\right)}\end{aligned} \tag{32}$$

Equation (32) indicates that P^1 is continuous and differentiable everywhere between breaking points.

APPENDIX C
PROOF OF PROPERTY 2

From *Property 1*, it is obvious $\frac{\partial P^1}{\partial \eta^1}$ is determined by $\mathcal{A}'$ and k, thus the breaking points of $\frac{\partial P^1}{\partial \eta^1}$ can be classified into two categories. From (15), with the increase of η^1 the number of positive η^h, i.e., k, will increase stepwise, and these breaking points are: $\alpha^k - \alpha^1, k = 1, 2, \ldots, H$, as shown in Fig. 7(a). The other type of breaking points are those can change $\mathcal{A}'$, i.e., the set of appliances whose power p_a^1 is smaller than the p_a^{max}. According to (13), p_a^1 decreases stepwise in η^1, thus the size of $\mathcal{A}'$ increases piecewise linearly. As shown in Fig. 7(b), let $\eta^1 = 0$ and $\mathcal{A}'(0)$ be the initial set of appliances whose power p_a^1 is smaller than the p_a^{max}. Then with the increase of η^1, $\mathcal{A}'$ keeps the same at the beginning and $|\mathcal{A}'|$ increases by 1 until for an appliance $a_1 \in \mathcal{A}/\mathcal{A}'(0)$, its power changes from $p_{a_1}^{max}$ to $W'^{-1}_{a_1}(\frac{\tau_{a_1} - \alpha^h - \eta^h}{o_{a_1}}) + \overline{p}_{a_1}$. That is, the breaking point associated with appliance a_1 is: $W'^{-1}_{a_1}(\frac{\tau_{a_1} - \alpha^h - \eta^h}{o_{a_1}}) + \overline{p}_{a_1} = p_{a_1}^{max}$, which gives

$$\eta_{a_1}^1 = \tau_{a_1}(0) - \alpha^1 - o_{a_1} W'_{a_1}\left(p_{a_1}^{max} - \overline{p}_{a_1}\right) \tag{33}$$

Similarly, we can obtain other breaking points $\eta_{a_2}^1, \eta_{a_3}^1, \ldots,$ for $a_2, a_3, \ldots, \in \mathcal{A}/\mathcal{A}'(0)$. Clearly, there are $|\mathcal{A}| - |\mathcal{A}'(0)|$ such breaking points, and without loss of generality, assume $\eta_{a_1}^1, \eta_{a_2}^1, \eta_{a_3}^1, \ldots,$ are in ascending order.

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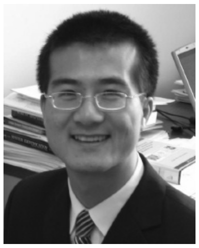

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